Step-by-Step Guide to Using test.py

This document explains how to use the **test.py** script for semantic segmentation inference and evaluation. The guide covers model loading, dataset setup, prediction generation, output visualization, and evaluation metrics.

1. Purpose of test.py

The test.py script is responsible for:

• Loading trained segmentation weights
• Running inference on validation or test images
• Generating prediction masks
• Calculating IoU metrics
• Saving visualization outputs

2. System Requirements

Recommended hardware and software:

• Python 3.10+
• NVIDIA GPU with CUDA support
• 16GB+ VRAM recommended
• Conda environment support

Clone Repository

```
git clone https://github.com/your-repo/falcon-segmentation.git
cd falcon-segmentation
```

Create Environment

```
conda create -n EDU python=3.10
conda activate EDU
```

Install Dependencies

```
pip install -r requirements.txt
```

Run test.py

```
python test.py
```

3. Dataset Structure

```
dataset/
■
■■■ Color_Images/
■
■■■ Segmentation/
```

4. What Happens During Execution

When test.py runs, the following operations occur: • Validation dataset loading • Image preprocessing • DINOv2 backbone loading • Segmentation head loading • Forward inference pass • Prediction mask generation • IoU calculation • Visualization saving • Metrics report generation

5. Generated Outputs

The script automatically generates: • masks/ → Raw segmentation masks • masks_color/ → Colored prediction masks • comparisons/ → Side-by-side comparisons • evaluation_metrics.txt • per_class_metrics.png

6. Important test.py Parameters

Argument	Description
--model_path	Path to trained .pth weights
--data_dir	Dataset directory
--output_dir	Where predictions are saved
--batch_size	Batch size for inference
--num_samples	Number of comparison images

7. Example Command

```
python test.py --model_path segmentation_head.pth --data_dir ./dataset --output_dir ./predictions --batch_size
```

8. Expected Console Output

```
Using device: cuda
Loading dataset...
Loading DINOv2 backbone...
Model loaded successfully!

Running evaluation...

Mean IoU: 0.9632

Prediction complete!
```

9. Troubleshooting

Common issues and fixes: • CUDA Out of Memory → Reduce batch size • Missing segmentation_head.pth → Verify model path • Dataset Not Found → Check dataset structure • Slow Inference → Ensure GPU acceleration is enabled

10. Final Notes

The test.py script is designed to evaluate trained semantic segmentation models on unseen Falcon offroad environments. It provides prediction outputs, evaluation metrics, and visualization results automatically. The generated masks and reports can be used directly for hackathon submission, analysis, and presentation purposes.

End of Report — test.py Usage Documentation